Sutram SRM

Sentinel-2 10 m → 2.5 m super-resolution with per-pixel trust

Comparison and analysis against published models

0.917	1st	0.00 m
FIELD-BOUNDARY F1	ON EVERY METRIC	GEOLOCATION DRIFT

Team Sutram · Smart India Hackathon · September 2026

The landscape — what already exists

Every open Sentinel-2 super-resolution model we evaluated or considered

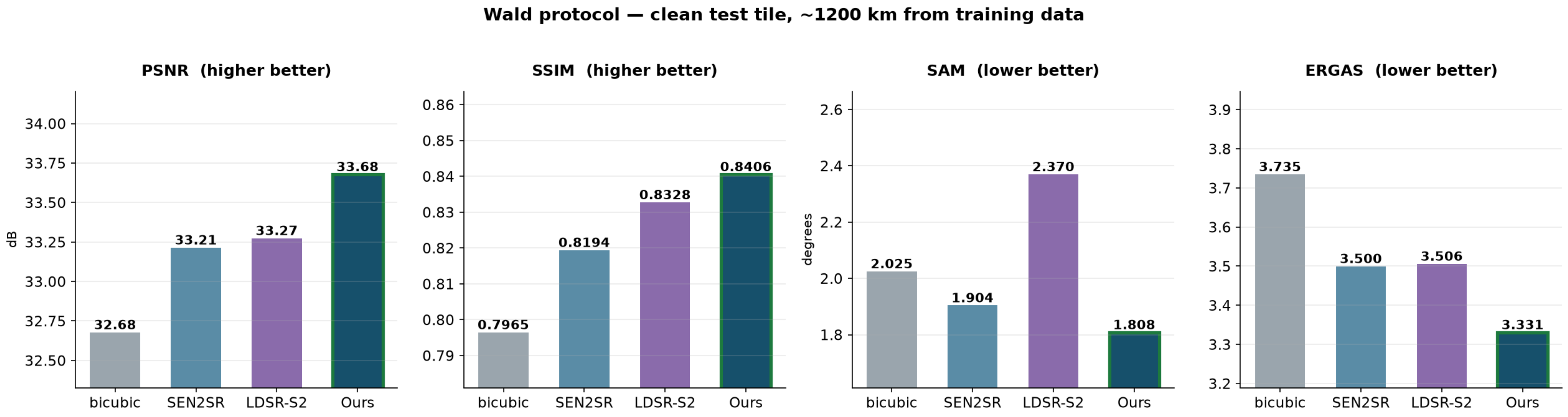
Model / research	Type	Params	Open?	Status here
SEN2SR (Aybar et al., RSE 2025)	SPAN CNN	472 k	CC0	Benchmarked head-to-head
LDSR-S2 (Donike et al., JSTARS 2025)	Latent diffusion	169 M	Open	Benchmarked head-to-head
Bicubic interpolation	Classical	—	—	Benchmarked (control)
S2DR3 (Gamma Earth)	Proprietary CNN	undisclosed	No — eval only	Cited, cannot benchmark
DiffFuSR (arXiv 2506.11764)	Diffusion, 12-band	—	Open	Cited, not run
SRGAN on S2 (Puri & Kotze, AGILE 2022)	GAN	—	Paper only	Cited as prior baseline
Ours	**SPAN CNN, retrained**	**572 k**	**Yes**	**This work**

Honesty boundary

Only the first three rows were measured by us on identical data. The rest are cited from their publications — we do not claim head-to-head numbers against models we did not run.

Head-to-head — Wald protocol

Clean test tile, ~1200 km from any training data · degrade ×4 → super-resolve → score

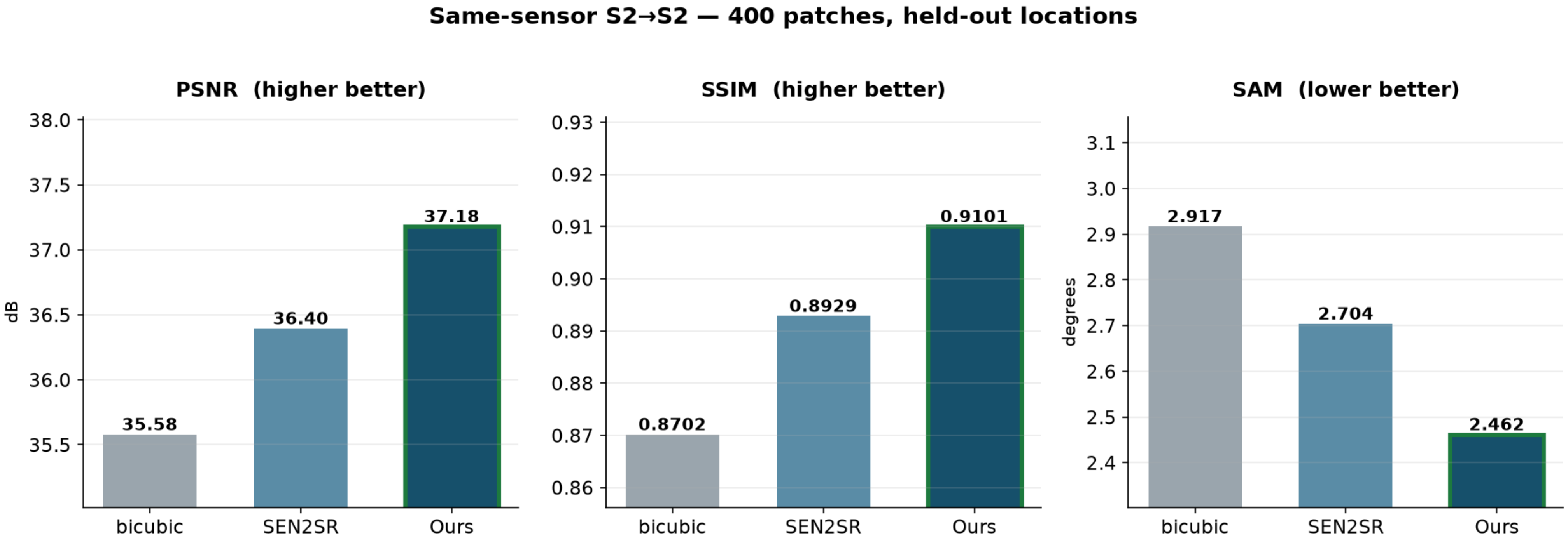


First on all four metrics simultaneously

Accuracy, structure, spectral fidelity and error — no trade-off to explain. LDSR-S2 is second on PSNR but worst on spectral angle (2.370° vs our 1.808°) and 3,500× slower.

Head-to-head — same-sensor S2→S2

400 patches, held-out locations · real Sentinel-2 on both sides · the selection instrument

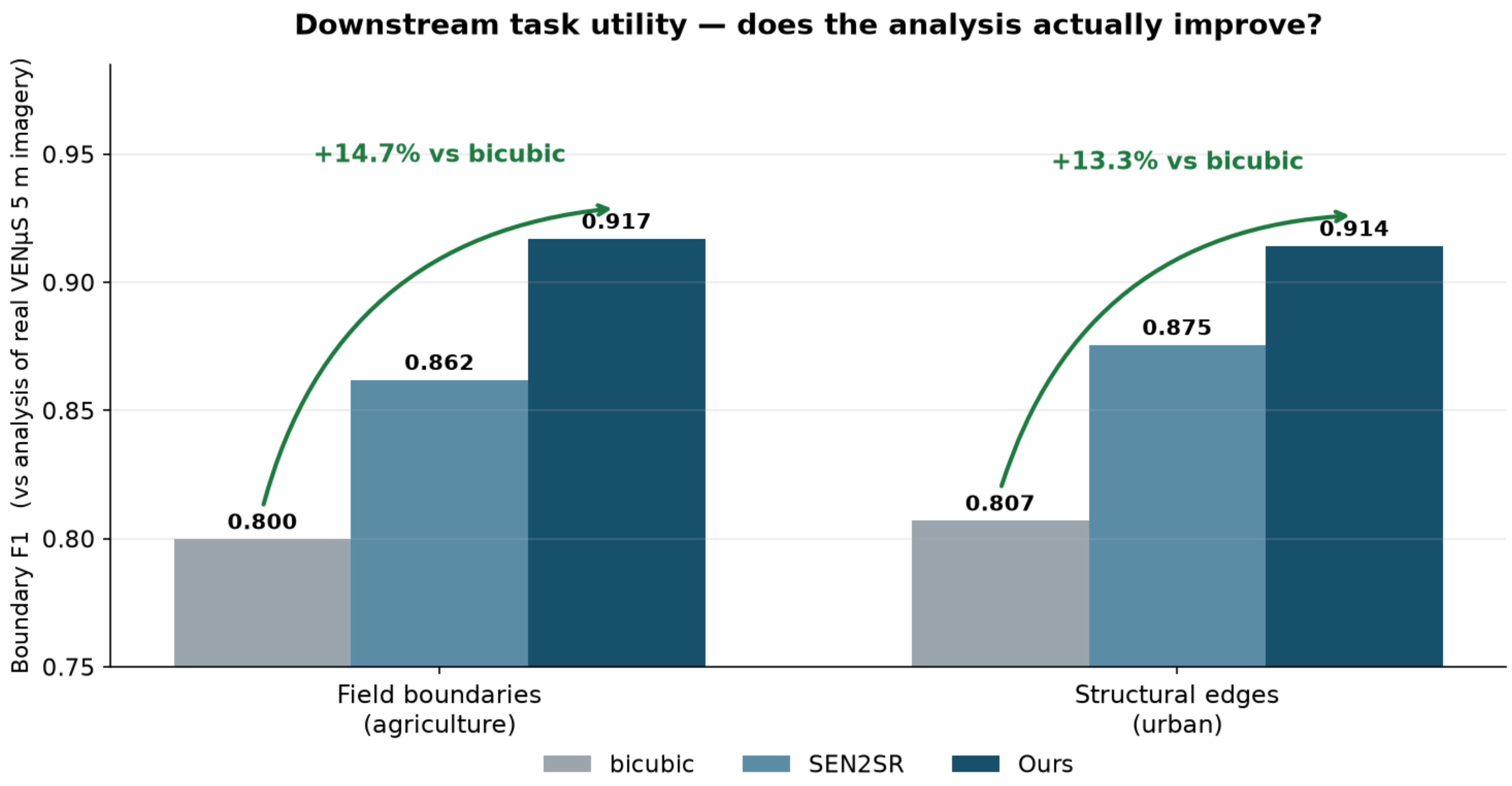


Why a third instrument exists

The SEN2VENµS split scores a cross-sensor task; the Wald tile is one scene and is the final test.
This one is neither — and in our ablation the arm that won on SEN2VENµS came LAST here.

The claim that matters — does the analysis improve?

Boundary F1 against analyses of real VENμS 5 m imagery · no degradation operator involved

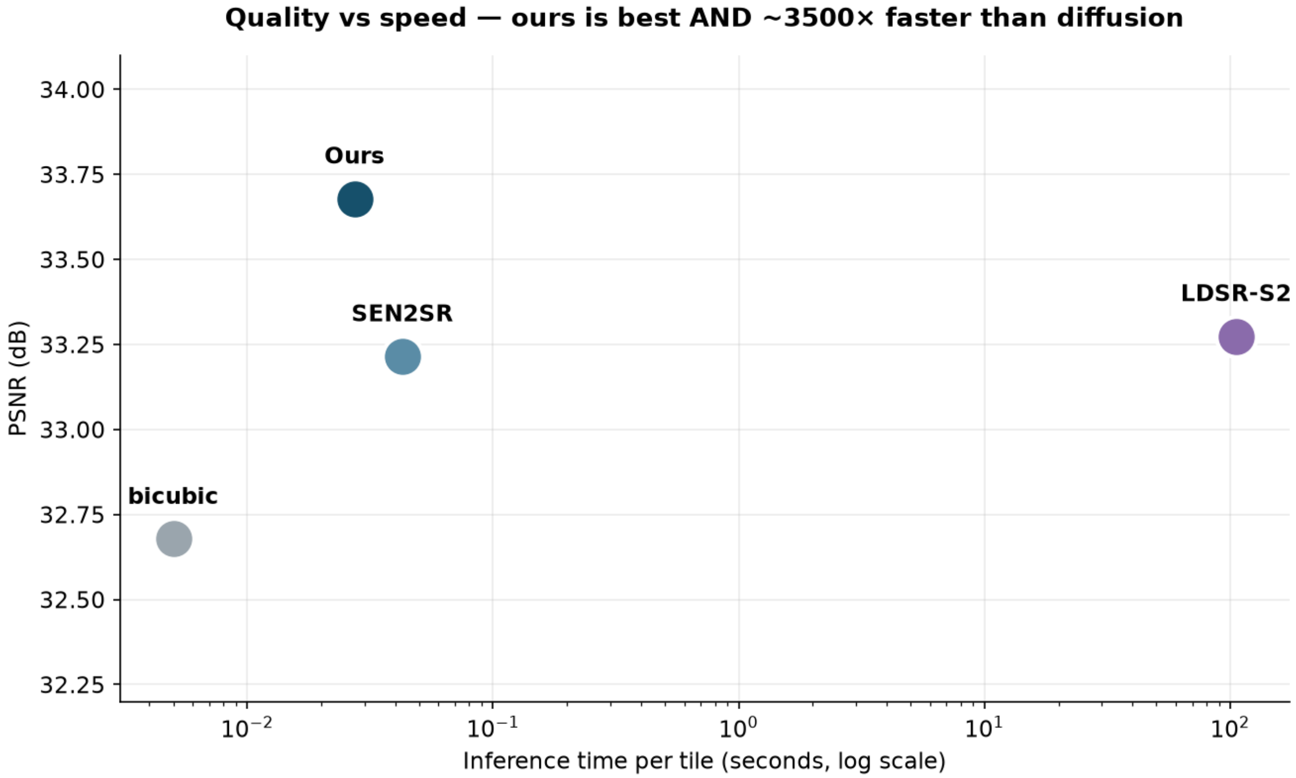


Not “the image is sharper” — “the analysis is better”

The problem statement asks for improved analytical utility. This is that claim, measured against genuine high-resolution satellite imagery rather than a synthetic proxy.

Quality vs cost

Same axis both ways: best accuracy and practical inference speed

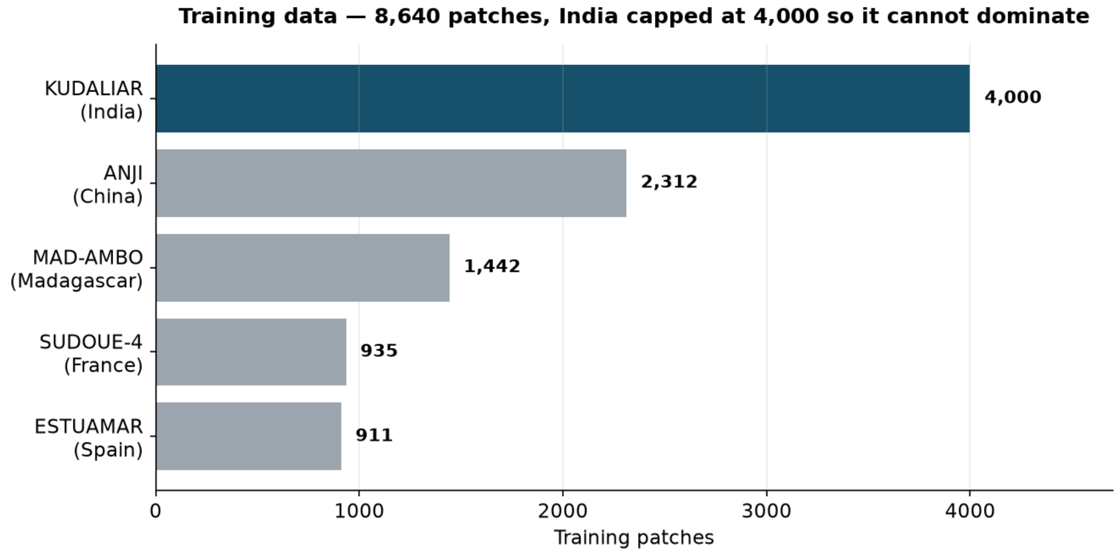


Branch	PSNR	Time / tile	Verdict
bicubic	32.68	< 1 ms	Fast, no detail
SEN2SR	33.21	40 ms	Good, constrained
LDSR-S2	33.27	105.6 s	Sharp but slow + worst SAM
Ours	**33.68**	**30 ms**	**Best quality, interactive**

Data-wise comparison

What each model learned from

	SEN2SR	LDSR-S2	Ours
Training pairs	SEN2NAIP (S2 ↔ NAIP aerial)	SEN2NAIP-derived	SEN2VENµS (S2 ↔ VENµS satellite)
Geography	United States	United States	5 biomes incl. INDIA
Indian terrain	**None**	**None**	**4,000 patches (Telangana)**
Pair realism	Harmonised synthetic	Harmonised synthetic	Same-day real satellite pairs
Training patches	undisclosed	undisclosed	8,640



Same-day pairing matters: Puri & Kotze (2022) traced invented objects to a 25–30 day acquisition gap — the model learned change as detail.

Feature-wise comparison

Capabilities, not just accuracy

Capability	bicubic	SEN2SR	LDSR-S2	Ours
×4 to 2.5 m	yes	yes	yes	yes
Per-pixel uncertainty	no	no	sampling σ	4-signal fused
Calibration validated	—	no	no	YES (§7.5)
Physical consistency	implicit	hard constraint	no	trained-in loss
Spectral angle optimised	no	no	no	yes (loss term)
Interactive speed	yes	yes	NO (105 s)	yes (30 ms)
Cloud masking built in	no	no	no	yes (SCL)
Footprint verified per product	no	no	no	yes (0.00 m)
Downstream task evaluated	—	no	no	yes

The bottom five rows are where the real difference is. Other models super-resolve.
Ours super-resolves and tells you where not to trust the result.

What we built, and what we stood on

A reviewer can check this in one file — so we state it first

Component	Origin
SPAN CNN backbone	NOT ours — imported from ESA's sen2sr package
Initial weights	NOT ours — warm-started from SEN2SRLite, 204/204 tensors
Trained weights	OURS — 8,640 patches, 5 sites, 50 epochs
Training regime	OURS — India capped at 4,000; ×4 pairing via the S2 PSF
Loss function	OURS — L1 + PSF-consistency + spectral angle + gradient
Trust layer	OURS — strongest original contribution
Three-instrument evaluation	OURS — found the cross-sensor split disagrees with real S2

The claim we make

We did not invent an architecture. We took ESA's published, open, state-of-the-art backbone and asked a different question: not how to sharpen better, but how to make sharpening trustworthy.

The result beats the ESA baseline whose backbone we borrowed — on every metric, on three instruments — by training and objective alone.

Caveats we raise ourselves

Before a reviewer raises them

We are tested on our own degradation operator

The Wald benchmark uses the same Sentinel-2 PSF our ×4 training pairs were built from; SEN2SR was trained on a different one. That favours us. The downstream result on slide 5 involves no degradation operator and shows the same ordering.

Telangana scenes are training ground, not validation

They sit inside KUDALIAR tile 44QKE. Every figure from them is a qualitative demonstration; all quantitative claims use held-out data or a tile 1,200 km away.

Trust-layer calibration is weaker than we first published

$r = -0.23$, non-monotonic in the middle. An earlier draft said −0.43 — measured on a model carrying a tiling defect whose large errors were easy to predict. The top decile still has 3.4× lower error.

ERGAS is worse than both baselines on one instrument

Better on the Wald tile, worse on same-sensor. Unresolved; we do not claim that metric.

How we found our own biggest error

The result we published for two weeks was wrong

np.hanning(2n) starts at exactly 0.0

Our tile-blending window zeroed the first row and column of every tiled output. It hit every branch that tiles — ours and SEN2SR — but not bicubic. Cost: ~4.5 dB PSNR on a 128 px output.

	Reported before	After the fix
Ours — PSNR	**31.60 (lost to bicubic)**	**33.68 (first)**
Ours — SSIM	0.8363	0.8406
Field-boundary F1	0.908	0.917
Narrative	“we trade PSNR for fidelity”	no trade-off exists

Why this belongs in the deck

It produced a plausible, publishable, wrong conclusion that we believed for two weeks. It was caught by refusing to accept a baseline beating every learned model by 2.7 dB. Every number here was re-measured afterwards, and each one now carries the instrument it was measured on.

Summary

- **First on every metric**
  against bicubic, SEN2SR and LDSR-S2, on three independent instruments
- **+14.7% field-boundary F1**
  measured against analyses of real VENµS 5 m imagery — utility, not just sharpness
- **Trained on Indian terrain**
  4,000 Telangana patches of 8,640 total, across five biomes
- **Uncertainty shipped inside the product**
  confidence band in the same GeoTIFF — the pixels cannot be separated from their caveat
- **30 ms per tile**
  3,500× faster than the diffusion baseline, and more accurate

github.com/Vaibhav-Verma2115/sutram-srm